

UNITE: Unified Network for Image and Text Embedding

Abstract—In this work, I introduce a novel image quality assessment model, UNITE, which leverages CLIP for enhanced performance. Additionally, the AIGN dataset is further enriched through the integration of LLaVA. Extensive experiments on the AIGN++ dataset demonstrate the superiority of my model, highlighting its effectiveness in comparison to existing methods.

Index Terms—AI-generated images, perceptual quality, image quality assessment

I. INTRODUCTION

AIGC (AI-generated content) has emerged as a transformative form of content creation, with significant advancements and applications across both academic and industrial sectors. Unlike traditional data, which is sourced from real-life experiences or human-generated content, AIGC is produced by AI systems that learn from extensive pre-existing datasets. Within the domain of AIGC, AI-generated images (AGI), particularly in the area of text-to-image generation, have seen considerable progress. A variety of AGI models have been proposed, resulting in substantial developments within the field.

Unlike images created by humans or photographs captured from the real world, AI-generated images are evaluated through a range of metrics, such as realism and human preference. Despite these advancements, there remains an urgent need for a reliable and standardized metric to evaluate the quality of AI-generated images. In AGI quality assessment (AGIQA), subjective evaluation across multiple dimensions is commonly employed. However, this approach faces several challenges, including high annotation costs, limited availability of annotations, and individual differences among evaluators.

Many methods used for image quality assessment tend to overlook the importance of contextual descriptions. However, additional descriptive information can enhance evaluation outcomes, particularly in assessing aspects such as rationality, technical quality, and naturalness. Therefore, it is crucial to develop a more comprehensive database and a multimodal input model to facilitate a more accurate and holistic assessment of AGI.

In this paper, I further enhance the AIGN database by creating AIGN++. Specifically, I have added descriptive labels and a three-dimensional evaluation score using LLaVA. Additionally, I have designed a novel image quality assessment model that supports both text and image inputs. Experimental results demonstrate that my model outperforms JOINT. I

also investigate the impact of different loss functions on the experimental outcomes.

In summary, the main contributions of this paper are as follows:

- I have developed AIGN++, an AI-generated image quality evaluation dataset based on AIGN, which includes evaluation descriptions and scores for images from both technical and natural dimensions.
- I proposed a novel image quality model, UNITE, which supports multimodal inputs from both images and text, and demonstrated its superior performance in experiments.
- I explored the impact of different loss functions on the performance of image quality evaluation tasks.

II. RELATED WORK

AIGC (Artificial Intelligence Generated Content). AIGC is an emerging form of content creation that has seen rapid advancements and significant applications across both academic research and industrial sectors. Unlike traditional content that is derived from real-world experiences or created by human authors, AIGC is generated by machine learning models, often based on large-scale datasets and sophisticated algorithms. This novel approach allows for automated and scalable content generation, which has sparked widespread interest due to its potential to enhance creativity and efficiency. However, as AIGC continues to evolve, it presents unique challenges in evaluating its quality, particularly in terms of the alignment between generated content (such as images) and accompanying textual descriptions.

AI-generated Image Assessment. Existing research on AI-generated image assessment primarily focuses on perceptual quality. Early objective metrics, such as the Inception Score (IS), assess perceptual quality by calculating the uniformity of AGI group features from the output of the Inception model. Distance-based methods, including the Fréchet Inception Distance (FID), Kernel Inception Distance (KID), and Precision-Recall, evaluate the discrepancy between the distributions of AGI and NSI. However, these methods are group-targeted and are not suitable for assessing individual images. Moreover, the widely used CLIPScore has reached a saturation point when comparing state-of-the-art generative models to authentic images and may inflate scores for models trained to optimize text-to-image alignment in the CLIP space. The limitations of these automatic measures have led to a

growing interest in human evaluations of perceived quality. Researchers such as Kirstain et al. have collected human preferences between two generated images from the same prompt for text-to-image tasks, while Wang et al. explored the impact of quality, fidelity, and correspondence of AI-generated images on human visual perception. Similarly, Li et al. conducted subjective evaluations to annotate images from both perceptual and alignment perspectives, with varying input prompts and internal parameters in AGI models. However, these studies often rely on coarse, single-voice, and overall subjective opinions, lacking a detailed exploration of specific factors and fine-grained, explainable evaluations for various generation tasks.

AGI Quality Database. Several databases have been constructed to facilitate the understanding of human visual preferences for AI-generated images based on text prompts, contributing to the field of AI-Generated Image Quality and Content-based Image Quality Assessment (AIGCIQA). Notable datasets include DiffusionDB, which contains a large number of images and prompts, AGIQA-1K, the first subjective database specifically designed for perceptual AGI quality assessment, and Pick-A-Pic. Additionally, HPS combines both perception and alignment evaluations with an overall score, while AGIQA-3K and AIGCIQA2023 provide fine-grained AGI quality assessments, addressing both perceptual quality and text-to-image (T2I) alignment. These databases have been essential for advancing the development of more precise, subjective evaluation methods for AI-generated images, and they serve as valuable benchmarks for assessing the performance of generative models across various tasks.

In addition to these, the AGIN database further enriches the field by offering a more detailed and large-scale collection of AI-generated images with human evaluations. This database consists of 6,049 AI-generated images and includes 907,350 ratings based on overall naturalness, as well as two key perspectives: technical quality and rationality. Each image is also annotated with the main influencing factors contributing to these scores. The AGIN database provides a deeper, fine-grained analysis of AI-generated image quality, specifically focusing on perceptual quality and alignment, which makes it a valuable addition for refining generative models and enhancing subjective evaluation methodologies in this area.

III. AGIN++ DATABASE

TABLE I
COMPARISONS BETWEEN THE AGIN++ AND OTHER AGIQA DATABASES

Database	Label Types
AGIQA-1k	Grades
AGIQA2023	Grades
AGIQA-3k	Grades
AGIN	Grades
AGIN++(Mine)	Grades, Evaluation

In this section, I will mainly introduce the AGIN++ dataset. Like AGIN, it contains 6,049 AI-generated images, but it also includes ratings for both naturalness and two sub-dimensions:

technical and rational. [11]My contribution lies in further enriching the dataset based on AGIN by adding two types of labels: rating and evaluation, as well as corresponding AI-generated scores for reference. A quick comparison can be made in Table 1.

For each image in the AGIN dataset, it is paired with a predefined prompt and input into the LLaVA model, with the output format strictly controlled. The required subsections of the output are selected, and the results are normalized and stored as corresponding labels. The prompt is designed to query three categories: technical, rational, and natural. [10]For each category, normalized evaluative text segments and quality scores are generated as labels. Figure 1 illustrates the design framework of our AGIN++ dataset.

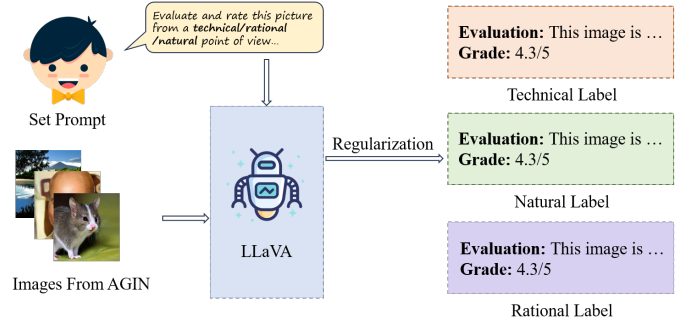


Fig. 1. Design of AGIN++ Database

IV. METHOD

CLIP is not the first model to be introduced for image quality evaluation tasks. [8] However, previous works heavily relied on large amounts of manually annotated data, making it challenging to transfer to new datasets, with significant limitations. Moreover, the training process requires considerable effort. [9]To address these challenges, I introduce large language models to assist in generating dataset labels and have designed a unique text encoder to enhance the model's ability to understand textual information. In this section, I will primarily introduce the model design of UNITE.

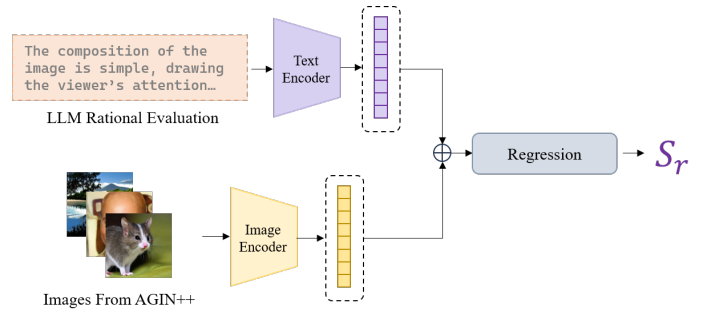


Fig. 2. Rationality Branch Framework

A. Overall Structure of UNITE

Following the relationships between the tech branch, ration branch, and natural branch discovered by AGIN, I generally adopted the model design approach of JOINT. For the tech branch, numerous research efforts have sought to localize the edited regions, which essentially involves training a model to identify systematic inconsistencies in generated images. [7] More recently, Zhang et al. further expanded this task to a fine-grained level, not only predicting the inpainted areas but also detecting and segmenting artifact regions that are perceptible to human observers. In this work, we utilize the detection model proposed in as the artifact extractor to guide patch partitioning. Note that the detection process itself is not the focus of this work. Given an image, the patch partitioning can be formulated as:

$$m, n = \text{AExtractor}(\mathcal{I}) \quad (1)$$

$$\mathcal{I}_{\text{rand}} = \text{RPart}\left(\mathcal{I}_{j \in [1, \frac{\mathcal{I}_H}{N}] \setminus \{m\}, k \in [1, \frac{\mathcal{I}_W}{N}] \setminus \{n\}}, N_8\right), \quad (2)$$

B. Text Encoder And Image Encoder

We utilize the text encoder $T(\cdot)$ and image encoder $V(\cdot)$ from CLIP. The text encoder is based on a Transformer architecture and is responsible for generating text representations from natural language. [3] In contrast, the image encoder is designed to map high-dimensional images into a low-dimensional embedding space. This encoder's architecture can resemble a Convolutional Neural Network (CNN), such as ResNet-50, or a Vision Transformer (ViT). In our setup, we employ these encoders separately to process our input textual information P and image information I , generating intermediate features that are then used to predict the quality score.

C. Image Quality Regression

To better adapt the CLIP features to the data distribution for the task of evaluating the quality scores of generated images, we follow the paradigm of general quality assessment tasks by employing a regression model $R(\cdot)$ to predict quality scores. [4] We concatenate the image features $F_i = V(I) \in \mathbb{R}^{1 \times N}$ and the textual features $F_p = T(P) \in \mathbb{R}^{6 \times N}$ to form the input features F , expressed as:

$$F = \text{concat}(F_i, F_p)$$

We process the concatenated features F through two fully connected (FC) layers. The parameters of the FC layers are learnable. The projection sizes are from 7×512 to 512, and then from 512 to 1. Finally, we obtain the predicted quality score S , formulated as:

$$S = R(F)$$

Throughout the entire learning process, we use the Mean Squared Error (MSE) as the loss function, defined as follows:

$$L = \frac{1}{N} \sum_{i=1}^N (S - y)^2$$

V. EXPERIMENTAL SETTINGS

A. Databases and Baselines.

I conduct experiments on our proposed AGIN database and select 5 state-of-the-art methods as references for comparison. These include JOINT and four contrastive language-image pre-training (CLIP) model-based IQA methods: CLIP-IQA, CLIP-IQA⁺, LIQE, and InternLM-XComposer. [2]

B. Implementation Details.

In the technical branch, we crop patches of size 64×64 , and the Swin Transformer is used as the backbone. The batch size is set to 64. All experiments are conducted under 5 train-test splits.

TABLE II
PERFORMANCE COMPARISON OF DIFFERENT METHODS

Methods	Technical		Rationality		Naturalness	
	SRCC↑	PLCC↑	SRCC↑	PLCC↑	SRCC↑	PLCC↑
CLIP-IQA	0.2114	0.3275	0.0348	0.0827	0.0167	0.1109
CLIP-IQA ⁺	0.4959	0.5595	0.2613	0.3189	0.3078	0.3550
LIQE	0.4928	0.5428	0.2457	0.2765	0.2974	0.3244
ILM-XCompose	0.4741	0.5085	0.3074	0.3271	0.3268	0.3566
JOINT	0.9558	0.9478	0.9217	0.9140	0.9070	0.9062
UNITE(Mine)	0.9561	0.9470	0.9531	0.9487	0.9201	0.9211

C. Ablation Study

For the JOINT model, I evaluated various loss functions, including PLCC, SRCC, KRCC, RMSE, and MAE. The mathematical formulations for these loss functions are as follows:

- 1) Pearson Linear Correlation Coefficient (PLCC):

$$\text{PLCC} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

- 2) Spearman Rank Correlation Coefficient (SRCC):

$$\text{SRCC} = 1 - \frac{6 \sum_{i=1}^n (r_i - s_i)^2}{n(n^2 - 1)}$$

where r_i and s_i are the ranks of x_i and y_i , respectively.

- 3) Kendall Rank Correlation Coefficient (KRCC):

$$\text{KRCC} = \frac{2(n_c - n_d)}{n(n - 1)}$$

where n_c and n_d represent the number of concordant and discordant pairs, respectively.

- 4) Root Mean Squared Error (RMSE):

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - y_i)^2}$$

- 5) Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |x_i - y_i|$$

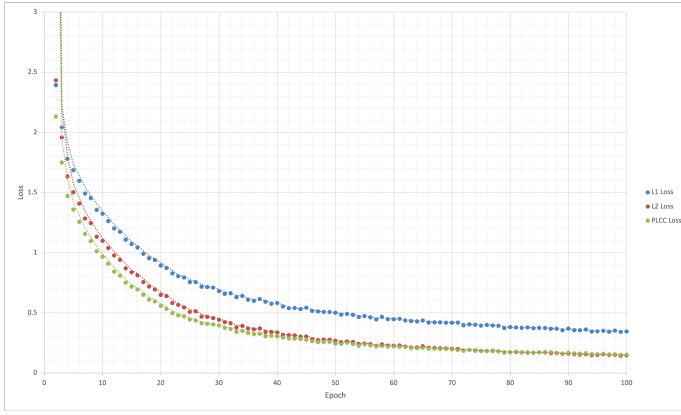


Fig. 3. Comparison of Different Loss Function

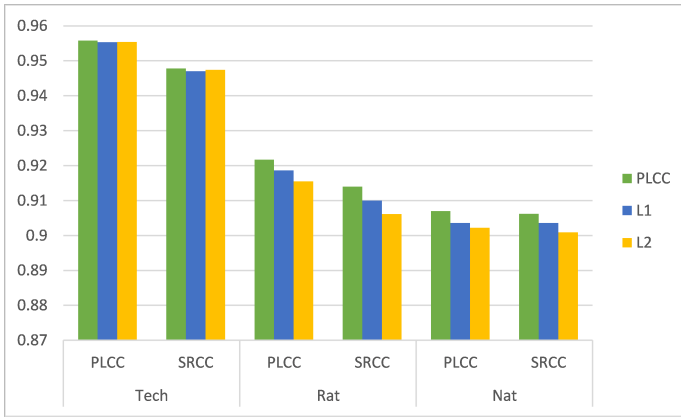


Fig. 4. Comparison of Different Loss Function

The observed differences in convergence speed and performance metrics (PLCC, SRCC, KRCC, RMSE, MAE, etc.) across various loss functions can be attributed to the following reasons:

1. Characteristics of Each Loss Function

Each loss function optimizes specific aspects of prediction quality: PLCC and SRCC: These correlation-based loss functions focus on capturing the relationship between predicted and ground truth rankings or values. As a result, they align well with evaluation metrics like PLCC and SRCC but often require more iterations to converge as the model adjusts to both global and local patterns in the data. L1 and L2 Losses (MAE and RMSE): These losses prioritize minimizing direct error magnitudes, which leads to faster convergence. However, their lack of direct emphasis on ranking consistency may result in slightly lower final performance on correlation-based metrics.

2. Convergence Speed

The convergence curves highlight that: L1 and L2 losses achieve faster initial reductions in error due to their emphasis on direct minimization of discrepancies. These functions are more sensitive to large initial errors, which accelerates early training. PLCC and SRCC losses converge more slowly because their optimization targets global ranking or correlation, requiring the model to develop a more nuanced understanding

of the data distribution. [1]

3. Performance Across Metrics

The final performance, as shown in the bar chart, varies depending on the alignment between the loss function and evaluation metrics: Models trained with PLCC and SRCC losses achieve higher performance on ranking-based metrics such as PLCC and SRCC, as these losses directly optimize these objectives. [5] L1 and L2 losses provide strong performance on magnitude-based metrics but slightly underperform in correlation-based evaluations due to their indirect optimization of ranking consistency.

4. Impact of Dataset Characteristics

The AGIN dataset spans technical, rationality, and naturalness domains, introducing variability in the relationship between input features and target labels: PLCC and SRCC losses adapt better to such variability, as they preserve ranking relationships across diverse domains. [6] L1 and L2 losses perform well when the data distribution is consistent and errors are uniformly distributed but struggle with complex domain-specific variations.

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